

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Problem - Solution Fit	DATE	July 2026

Problem - Solution Fit

Problem

Manual credit card application review is slow, inconsistent, and error-prone, especially at the scale banks operate.

Solution

A trained classification model that evaluates applicant financial and demographic data automatically, integrated into a web application for instant predictions.

Fit Analysis

Problem Aspect	How the Solution Addresses It
Slow manual review	Instant model prediction replaces manual cross-checking
Inconsistent decisions	A single trained model applies the same criteria to every applicant
Human error at scale	Automated feature-based evaluation using historical outcomes
Lack of applicant transparency	Self-service eligibility check before formal application

Validation

The selected model (Random Forest) was validated using a held-out test set, achieving an F1-score of 0.44 for the rejected class and 84% overall accuracy, confirming the solution meaningfully distinguishes approved from rejected profiles rather than defaulting to the majority class.